[COLLEGE / UNIVERSITY NAME PLACEHOLDER]

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING / INFORMATION TECHNOLOGY B.Tech Second Year | Course: Internet Programming Laboratory [Subject Code: _____]

ASSIGNMENT REPORT: CINEBOOK – MOVIE TICKET BOOKING WEBSITE

Academic Session: 2026–2027 **Faculty Guide:** [Faculty Name / Designation] **Project Type:** Full-Stack Group Web Application (5 Members) **Submission Date:** November 2026

1. Introduction & Objectives

CineBook is a full-stack, responsive movie ticket booking web application developed to satisfy the curriculum objectives of the B.Tech Internet Programming syllabus. The application bridges client-side interactive DOM interfaces with an asynchronous Node.js/Express REST backend and a relational MySQL database. The primary goals are: (a) enabling users to securely register, authenticate, and reserve cinema tickets with interactive seat selection; (b) validating transaction data and preventing double-booking; and (c) providing an administrative CRUD module to manage cinema catalogs.

2. Technologies Used

● **Frontend:** HTML5 Semantic Elements, CSS3 (Flexbox, Grid, Variables, @media print), Vanilla JavaScript (ES6+), DOM Manipulation, Fetch API. ● **Backend:** Node.js runtime, Express.js REST server, express-session (cookie session state), bcryptjs (password hashing), dotenv. ● **Database:** MySQL relational database, mysql2 promise pool, parameterized SQL queries preventing SQL Injection.

3. Explanation of the Five Connected Webpages & Workflow

● **Workflow:** Movie Selection → Theater Selection → Date & Showtime → Seat Matrix → Confirmation Ticket. **1. Page 1 – Registration (signup.html):** Validates full name, email regex, password strength, and confirmation equality. Stores accounts using salted bcrypt hashes on the server to prevent duplicate email registrations. **2. Page 2 – Login (login.html):** Authenticates credentials against stored password hashes. Establishes express-session cookie state and redirects users back to their pending booking destination. **3. Page 3 – Movie Browsing (movies.html):** Dynamically queries GET /api/movies and constructs DOM cards with titles, genres, durations, and pricing. Features live client-side search and genre filtering without page reloads. **4. Page 4 – Theater & Seat Booking (booking.html):** Allows selecting cinema theaters dynamically (IMAX Laser, PVR Premiere, Cinepolis 4DX, Royale VIP), show date, showtime, and interactive 5x8 seat grid. Checks theater-specific seat occupancy, prevents duplicate bookings in the same hall, and recalculates totals on the server. **5. Page 5 – Ticket Confirmation & Printing (confirmation.html):** Fetches verified booking details from GET /api/booking/:id with ownership protection. Displays cinema ticket with theater name, screen, location, barcode, and triggers browser printing via JavaScript window.print() formatted with @media print rules. **6. Module IV Bonus – Admin CRUD (admin.html):** Role-protected dashboard allowing administrators to Create, Read, Update, and Delete movies and monitor bookings.

4. MySQL Database Schema Explanation

The relational database "cinebook" is structured into four normalized tables with foreign key constraints: ● **users table:** user_id (INT, PK, AI), name (VARCHAR), email (VARCHAR, UNIQUE), password (VARCHAR, bcrypt hash), role (VARCHAR, default "user"). ● **movies table:** movie_id (INT, PK, AI), title (VARCHAR), description (TEXT), genre (VARCHAR), duration (INT min), price (DECIMAL), poster (VARCHAR). ● **theaters table:** theater_id (INT, PK, AI), theater_name (VARCHAR), location (VARCHAR), screen_name (VARCHAR). ● **bookings table:** booking_id (INT, PK, AI), user_id (FK to users), movie_id (FK to movies), theater_id (FK to theaters), show_date (DATE), show_time (VARCHAR), seats (VARCHAR), ticket_count (INT), total_price (DECIMAL), created_at (TIMESTAMP).

5. Individual Contributions of All Five Group Members

Member	Student Name / Roll	Assigned Page	Implemented Features & Syllabus Concepts
Member 1	[Student 1 Name] [Roll Number 1]	Page 1: Registration (signup.html)	Form validation (regex, length, equality), Fetch POST /api/signup, bcrypt hashing, error alert handling, redirect.
Member 2	[Student 2 Name] [Roll Number 2]	Page 2: Login (login.html)	Authentication via POST /api/login, express-session state, HTTP-only cookie, redirect logic, feedback banners.
Member 3	[Student 3 Name] [Roll Number 3]	Page 3: Movie Catalog (movies.html)	Dynamic card DOM creation from GET /api/movies, client search and genre filtering, responsive grid styling.
Member 4	[Student 4 Name] [Roll Number 4]	Page 4: Seat Booking (booking.html)	Interactive 5x8 seat map, booked seat detection, client-side dynamic pricing, login check guard, server revalidation.
Member 5	[Student 5 Name] [Roll Number 5]	Page 5: Ticket Output (confirmation.html)	Booking retrieval via GET /api/booking/:id, ticket layout, window.print() integration, @media print formatting.

6. Conclusion

The CineBook project successfully fulfills all curriculum requirements for the B.Tech Internet Programming laboratory coursework. By combining clean semantic HTML5, modern CSS3 styling, vanilla JavaScript DOM interactivity, Node.js/Express

REST APIs, and a relational MySQL database, the project demonstrates real-world software engineering practices. All 16 automated backend test assertions passed, guaranteeing security, price integrity, and double-booking prevention. The codebase is modular, clean, and easily explainable in an academic viva.

Student Signatures: (1) _____ (2) _____ (3) _____ (4) _____ (5) _____ Faculty Examiner Signature: _____ Date: _____